Sean Luka Girgis

Senior AWS Data Engineer | Data Architecture, SQL Reporting & Cloud Migration

214-315-2190 | seanlgirgis@gmail.com

LinkedIn | GitHub | Portfolio

Professional Summary

Senior AWS Data Engineer and data architecture specialist with 20+ years of enterprise technology experience across cloud data platforms, SQL reporting layers, semi-structured data processing, schema design, ETL/ELT pipelines, metadata-driven reporting, and stakeholder-facing analytics. Strong background using Python, SQL, AWS S3, Glue, Athena, Redshift, Oracle, PySpark, XML, JSON, performance tuning, documentation, and production monitoring to modernize data workflows and make complex data queryable for business users. Experienced translating ambiguous reporting needs into durable data models, migration plans, architecture decisions, and operational runbooks.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable reporting, analytics, and forecasting workloads.
- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for business and technical stakeholders.
- Migrated heavy forecasting and analytics workloads from on-prem Oracle patterns toward AWS/Redshift-backed processing, improving scalability and enabling cloud-based reporting workflows.
- Designed SQL reporting structures, historical retention schemas, and queryable data layers that made complex infrastructure telemetry easier to analyze and consume.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, partitioning patterns, retention strategy, and reporting data models for performance, maintainability, and cost-aware access.
- Built metadata-aware reporting and documentation practices, including metric definitions, pipeline logic, operational support steps, and data interpretation notes for technical and non-technical audiences.
- Validated and reconciled data across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve trust, completeness, and data quality.
- Monitored production data workflows, investigated failures, improved operational runbooks, and supported issue resolution across high-volume enterprise data pipelines.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and hospitality systems.
- Led the FAST project, mining real-user and synthetic performance data to identify optimization opportunities and communicate findings to engineering teams.
- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and data/reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting reporting, monitoring, and operational analytics.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed dashboards, alert logic, SLA thresholds, technical documentation, and reusable reporting standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to interpret data, troubleshoot bottlenecks, and improve production visibility.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, design findings, and release-readiness thresholds used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and reporting processes for telecom-scale transaction systems.
- Worked with XML, SQL, Oracle, PL/SQL, messaging, and integration interfaces supporting billing, customer-management, and operational reporting workflows.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.
- Built automation scripts in Korn shell, Perl, Awk, Syncsort, and SQL to support data processing, reporting, deployments, and operational housekeeping.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, and reporting access.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine
Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, and deliver capacity insights.
Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit

AI-Powered Job Search Pipeline
Built a Python automation pipeline with semantic duplicate detection, LLM-based job scoring, JSON artifact generation, quality gates, document rendering, and application tracking using FAISS, sentence-transformers, and LLM APIs.
Technologies: Python, FAISS, Sentence Transformers, LLM APIs, JSON, Automation